Source-Based AI Recommendation Explanations — Analysis Plan v1

1. Analysis Objectives

The analysis will examine how the information source disclosed in an AI recommendation explanation affects perceived relevance, perceived source credibility, situational privacy concern, trust, and reliance. The analysis will first evaluate the quality of the measures and experimental manipulation, then test the pre-specified directional hypotheses, and finally examine indirect effects and exploratory comparisons involving the mixed-source condition.

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2. Data Preparation

Each participant will contribute one row to the analysis dataset. The dataset will include experimental condition, responses to all multi-item measures, behavioral reliance, manipulation checks, and demographic/control variables. Before hypothesis testing, the data will be checked for missing responses, duplicate records, invalid condition coding, and other obvious data-quality problems.

Any exclusion criteria used for the pilot or main study will be specified before substantive hypothesis testing.

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3. Scale Construction

Multi-item constructs will be scored by averaging their corresponding items after checking that the items demonstrate acceptable internal consistency.

- Perceived Relevance = mean(PR1, PR2, PR3)
- Source Credibility = mean(SC1, SC2, SC3)
- Privacy Concern = mean(PC1, PC2, PC3)
- Trust = mean(TR1, TR2, TR3, TR4)
- Reliance Intention = mean(RI1, RI2, RI3)

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4. Reliability and Measurement Checks

Internal consistency will be examined for each multi-item construct before creating the final composite scores. Cronbach's alpha and/or McDonald's omega may be used as reliability indicators. Item-level patterns will also be inspected to identify items that behave inconsistently with the rest of the scale.

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5. Manipulation Checks

Manipulation-check responses will be examined to determine whether participants perceived the explanation-source manipulation as intended.

- Behavior-Based: higher MC1
- Review-Based: higher MC2
- Product-Feature-Based: higher MC3
- Mixed-Source: higher MC1, MC2, MC3, and MC4
- Control: generally lower source-specific ratings

Failure of the manipulation checks would be interpreted as a stimulus-design problem rather than automatically as evidence against the theoretical hypotheses.

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6. H1–H3: Experimental Condition Effects

One-way ANOVA models will first be used to evaluate overall condition differences. The theoretically specified hypotheses will then be evaluated using planned contrasts.

H1a:

- C2 Behavior-Based vs C1 Control
- DV: Perceived Relevance

H1b:

- C4 Product-Feature-Based vs C1 Control
- DV: Perceived Relevance

H2:

- C2 Behavior-Based vs C1 Control
- C2 Behavior-Based vs C3 Review-Based
- C2 Behavior-Based vs C4 Product-Feature-Based
- DV: Privacy Concern

H3:

- C3 Review-Based vs C1 Control
- DV: Source Credibility

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7. H4: Psychological Predictors of Trust

H4 will be evaluated using multiple linear regression, with Trust as the dependent variable and Perceived Relevance, Perceived Source Credibility, and Privacy Concern as predictors.

Trust = β0 + β1(Relevance) + β2(Credibility) + β3(Privacy) + error

β1 > 0, β2 > 0, β3 < 0

Because the psychological mediators are measured rather than experimentally randomized, these regression relationships will be interpreted primarily as associations rather than standalone causal effects.

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8. H5: Trust and Reliance

H5 will be evaluated using linear regression with Reliance Intention as the dependent variable and Trust as the primary predictor.

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9. Behavioral Reliance Analysis

Behavioral reliance will be analyzed separately from reliance intention because it is a binary outcome. Logistic regression will be used to examine whether Trust and/or experimental condition predict the probability of accepting the AI-recommended product.

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10. H6: Indirect Effects

After the simpler condition-effect and regression models are evaluated, indirect effects will be examined using mediation/path analysis.

- Behavior-Based → Relevance → Trust → Reliance
- Behavior-Based → Privacy Concern → Trust → Reliance
- Review-Based → Source Credibility → Trust → Reliance
- Product-Feature-Based → Relevance → Trust → Reliance

Bootstrap confidence intervals will be used to evaluate indirect effects.

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11. Mixed-Source Exploratory Analyses

The Mixed-Source condition will be treated as exploratory in the initial study. It will be compared with the control and single-source conditions on perceived relevance, source credibility, privacy concern, trust, and reliance. No directional hypothesis will be claimed for these comparisons.

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12. Interpretation Principles

1. Statistical significance will not be interpreted as evidence that an effect is practically large; effect sizes and confidence intervals will also be considered.
2. A non-significant difference will not be interpreted as proof that two conditions are equivalent.
3. Associations among measured psychological variables will not automatically be interpreted as causal relationships.
4. Pilot results will primarily be used to evaluate manipulation quality, measurement performance, and experiment feasibility rather than to make substantive claims about consumer behavior.